

Tide retrospective: the pencil identity for pairs of potentials

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-08

1 Setting

This tide opens a new direction for the laplace seabed: statements about *pairs* of potentials, toward the identifiability theorem of the germbij note (“What expectation values know about the loss landscape”, SRI, August 2026). That note proves that for nonnegative real analytic losses L_1, L_2 with common compact zero locus, equality of the asymptotic expansions of $\int \phi e^{-tL_i}$ for every observable ϕ forces $L_1 = L_2$ near the locus. The proof has three elementary ingredients: an exact *pencil identity* converting the difference of Boltzmann integrals into an average over the interpolating family $L_s = (1-s)L_1 + sL_2$; the choice of observable $\phi = g\psi$ with $g = L_2 - L_1$, which makes the integrand positive; and a sector lower bound producing a polynomial rate that contradicts $o(t^{-\infty})$. Everything in the seabed so far concerns a single potential; the pencil identity is the minimal first step outward, and it is the ingredient the other two compose with.

2 The thing formalised

The pencil identity, in pointwise and integrated form, together with the comparison inequality and the partition-function specialisation. For $L_1, L_2 : \mathbb{R} \rightarrow \mathbb{R}$, $g = L_2 - L_1$ and $L_s = L_1 + s g$:

$$e^{-tL_1(w)} - e^{-tL_2(w)} = t \int_0^1 g(w) e^{-tL_s(w)} ds, \quad \int_{\mathbb{R}} \phi (e^{-tL_1} - e^{-tL_2}) dw = t \int_0^1 \int_{\mathbb{R}} \phi g e^{-tL_s} dw ds,$$

the second under an explicit product-integrability hypothesis; and for $0 \leq s \leq 1$, $t \geq 0$ and pointwise nonnegative potentials, the comparison $e^{-t(L_1+L_2)(w)} \leq e^{-tL_s(w)}$. The Lean signatures (file `Laplace/Pencil.lean`, all sorry-free):

```
theorem exp_pencil_identity (L1 L2 : R -> R) (t w : R) :
  exp (-(t * L1 w)) - exp (-(t * L2 w))
  = t * intervalIntegral (fun s =>
    (L2 w - L1 w) * exp (-(t * (L1 w + s * (L2 w - L1 w))))) 0 1

theorem pencil_identity_integrated (L1 L2 phi : R -> R) (t : R)
  (hint : Integrable (uncurry ...) ((volume.restrict (Ioc 0 1)).prod
    volume)) :
  integral (fun w => phi w * (exp (-(t * L1 w)) - exp (-(t * L2 w))))
  = t * intervalIntegral (fun s => integral (fun w =>
    phi w * ((L2 w - L1 w) * exp (-(t * (L1 w + s * (L2 w - L1 w)))))
  ))) 0 1
```

```
lemma exp_pencil_ge ... : exp (-(t * (L1 w + L2 w))) <= exp (-(t * (L1 w +
s * (L2 w - L1 w))))
```

```
theorem partitionFunction_pencil ... :
  partitionFunction L1 t - partitionFunction L2 t = t * ...
```

(Signatures abridged; the source file has the precise forms, with the project's Boltzmann convention `exp (-(t * L x))` shared with `Laplace/Gibbs.lean`.)

3 Proof strategy

The pointwise identity is the fundamental theorem of calculus for $s \mapsto e^{-t(a+sg)}$ on $[0, 1]$: the derivative is $-(tg) e^{-t(a+sg)}$ by the chain rule combinators on `HasDerivAt`, the integrand is continuous hence interval integrable (discharged by `fun_prop`), and the `hasDerivAt` form of the interval FTC¹ gives $\int_0^1 = F(1) - F(0)$; a `calc` block moves the sign and the constant t through the interval integral.

The integrated form is the pointwise identity under the outer integral (`simp only [key]` with the pointwise identity as a rewrite family), the constant t pulled out, and one application of Fubini for a product with one interval factor,² whose hypothesis is exactly the product-integrability assumption of the theorem after rewriting `uLoc01 = Loc01 (Set.uLoc_of_le)`. The comparison lemma is monotonicity of `exp` plus a `nlinarith` certificate for $s(L_2 - L_1) \leq L_2$ given $0 \leq s \leq 1$ and nonnegativity, with the two product hints $s \cdot L_1 \geq 0$ and $(1 - s) \cdot L_2 \geq 0$ supplied explicitly. The partition-function corollary is the $\phi = 1$ case plus `MeasureTheory.integral_sub`.

4 Roadblocks and resolutions

This tide was unusually smooth; three small snags, all local. First, a `simp`-normalisation mismatch in the chain-rule step: the term `(h.const_mul t).neg` already has exactly the right type, and wrapping it in `simp [mul_add, neg_add]` rewrote the function to a shape the goal did not have. Removing the `simp` fixed it; the lesson (known from earlier tides) is to try `exact` before normalising. Second, the Mathlib style linter rejects `(0:R)` for `(0 : R)`; a one-shot `sed` conformed the file. Third, in the $\phi = 1$ specialisation the hypothesis transport under `Function.uncurry` does not follow from `simp [one_mul]` (`simp` will not rewrite under the `uncurry` without unfolding it); an explicit `funext` equality of the two uncurried functions, transported with $\blacktriangleright$, was needed.

Operationally, the deliberation consult was the slowest step: the API call hung for several hours before returning (a known failure mode of the query helper), and the formalisation was completed while it ran. The response, when it arrived, endorsed the pencil identity as the minimal target and the exact architecture that had been built, and its one improvement (state the identity first in scalar form, derive the functional version as a one-line corollary) was adopted in a final refactor. The verdict and the vote are recorded in the tide log alongside this file.³

¹`intervalIntegral.integral_eq_sub_of_hasDerivAt`.

²`MeasureTheory.intervalIntegral.integral_swap`.

³Tide log: `tide-log/2026-08-08-19-11-tide-germbij-pencil.md`; GPT consult saved verbatim as `gpt55-germbij-pencil.v1.md` in the same directory.

5 What was Mathlib, what was new

Everything load-bearing was Mathlib lookup: the interval FTC, the Fubini swap for a product with one interval factor, the `HasDerivAt` chain-rule combinators, and elementary rearrangements of integrals. New content is the four statements themselves and their composition; no new infrastructure, no new integration theory. That is as it should be for a first step outward: the mathematical novelty of the germbij chain is concentrated in the sector bound and the final contradiction, deliberately left for the next tides.

6 Lessons

The pencil identity confirms that pairs-of-potentials statements sit comfortably on the existing seabed: the Boltzmann convention of `Gibbs.lean` composes with interval integration over the pencil parameter without any impedance mismatch. Two workflow lessons, both reinforcing existing `CLAUDE.md` guidance: prefer `exact` over `simp` when a combinator already has the target type; and hypothesis transport under `Function.uncurry` wants explicit `funext` rather than `simp` lemmas about the uncurried arguments.

7 Follow-ups

- **Sector lower bound (next tide).** The 1D version with factored hypotheses: $K \geq 0$, $K \leq C_0 w^2$ and $|a| \geq c w^m$ on $[0, r_0]$ give $\int_{t^{-1/2}}^{2t^{-1/2}} a^2 e^{-tK} \geq c^2 e^{-4C_0} t^{-m-1/2}$. Elementary, via the constant lower bound for set integrals;⁴ the numerical check is already in the tide log.
- **The identifiability contradiction.** Composing the integrated pencil identity, the comparison lemma `exp_pencil_ge`, and the sector bound into the 1D version of germbij Theorem 7.3, with the analytic-germ input taken as the factored hypothesis $|g| \geq c w^m$.
- **Integrability bookkeeping.** A small API lemma producing the product-integrability hypothesis of `pencil_identity_integrated` from continuity plus compact support of ϕ , to make the theorem easy to apply downstream.

⁴`MeasureTheory.setIntegral_ge_of_const_le`.